

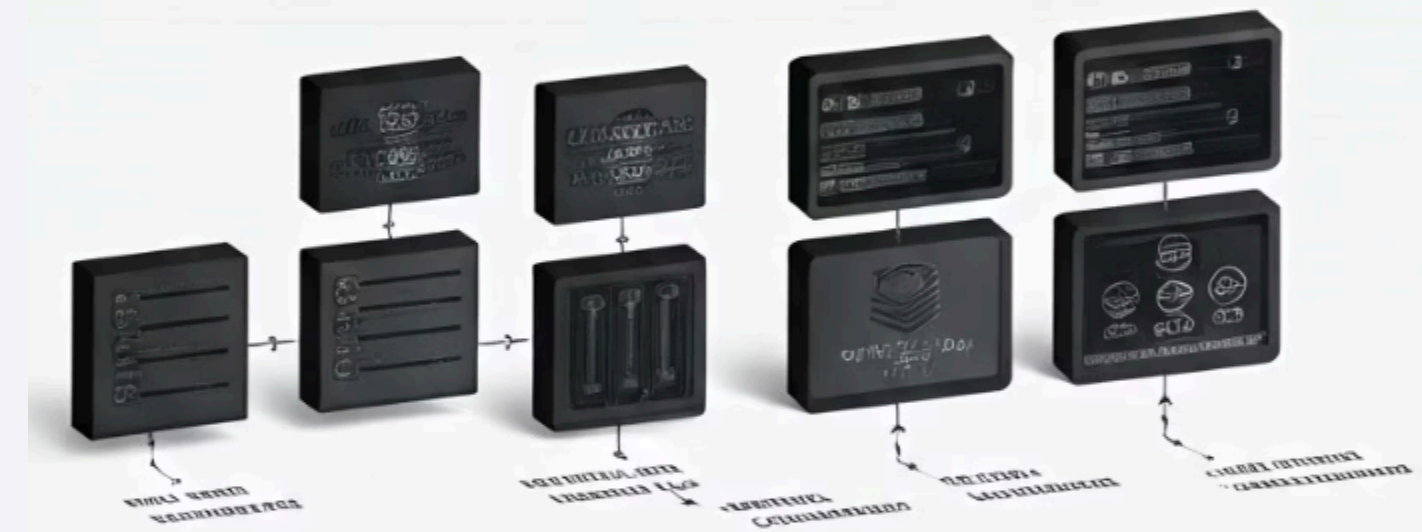


Real-Time data Analysis System

Python-Powered Stock Market Dashboard

System Overview

The purpose of this system is to provide real-time monitoring and analysis of stock market data. The core components include data acquisition from the Alpha Vantage API, a statistical analysis engine, and an interactive web dashboard.



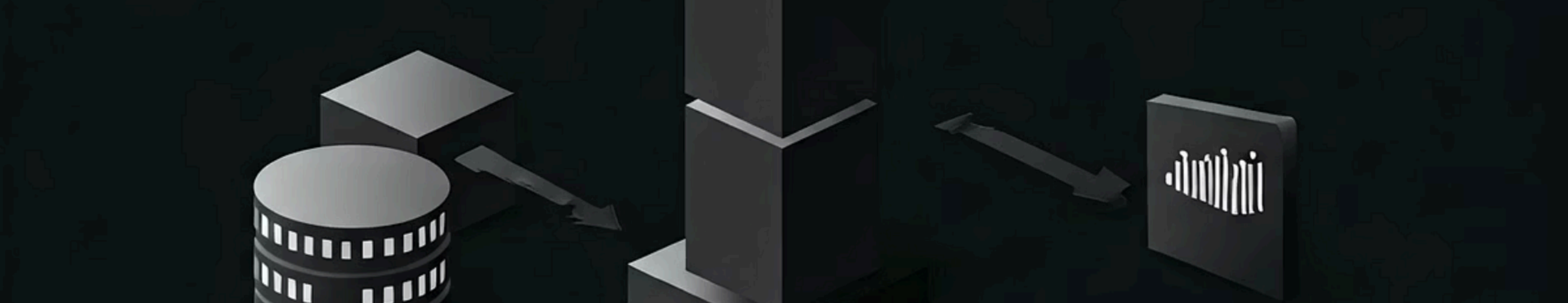
Technical Architecture

Backend

The backend is built using Python with multi-threading capabilities. The data flow follows the sequence: API → Queue → Analyzer → Web Server → Browser.

Key Libraries

- Requests, NumPy, Threading
- HTTP.Server, Socketserver
- Plotly.js (frontend)



Data Pipeline

1 Data Source

The system pulls data from the Alpha Vantage API, with a fallback mechanism in place.

3 Storage

In-memory data structures with thread synchronization are used for storage.

2 Processing

The data is subjected to statistical analysis and anomaly detection.

4 Tracked Stocks

The system currently tracks the following stocks: AAPL, GOOGL, MSFT, AMZN, META.

Core Features



Real-time Data Updates

The system updates stock data at 15-second intervals.



Multi-Stock Tracking

Users can track and compare multiple stocks simultaneously.



Statistical Analysis

The system provides real-time calculations of mean and standard deviation.



Responsive Dashboard

The web-based dashboard adapts to various device sizes and resolutions.

Dashboard Overview

Interactive Charts

- Price trends (time series)
- Volume comparison (logarithmic scale)

Real-time Statistics

- Current prices and volumes
- Statistical indicators
- Anomaly warnings

Data Analysis Techniques

1 Statistical Methods

- Rolling averages and standard deviations
- Z-score anomaly detection (threshold: 2.0)

2 Resilience Features

- API fallback with realistic simulated data
- Error handling and logging

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Technical Highlights

Multi-threaded Architecture

- Producer-consumer pattern with thread-safe queue
- Concurrent data processing and web serving

Web Technologies

- Python's built-in HTTP server
- Client-side JavaScript for dynamic updates
- Plotly.js for interactive visualization

Live Demo

You can access the live dashboard at <http://localhost:8051>. The main features to observe include real-time price and volume updates, statistical analysis cards, and anomaly indicators.

